Deep Learning Final Project

Pneumonia Classification from Chest X-Ray Images

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1 Introduction

1.1 Project Goal

This project addresses the task of **multi-class classification** of chest X-ray images into three categories:

1. **Normal** – healthy lungs
2. **Viral Pneumonia** – pneumonia associated with viral infection
3. **Bacterial Pneumonia** – pneumonia associated with bacterial infection

The main challenge lies in the subtle visual differences between viral and bacterial cases, as well as the imbalance between classes.

1.2 Dataset

The dataset was obtained from Kaggle (Chest X-Ray Images – Pneumonia):
`https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia`

After reorganizing the dataset into three classes (Normal / Viral / Bacterial), the following distribution was used:

- **Total images:** 5,863

1.3 Class Distribution

  - Normal: 1,583 images
  - Viral Pneumonia: 2,780 images
  - Bacterial Pneumonia: 1,500 images

- **Train/Validation split:** 80/20 split from the training folder using `validation_split=0.2`

1.4 Motivation

Automated pneumonia detection from chest X-rays can support clinicians by enabling faster screening and prioritization of suspected cases, especially in resource-limited or high-load settings. In addition, distinguishing between **bacterial** and **viral** pneumonia is clinically important because treatment strategies differ (e.g., antibiotics are typically relevant for bacterial infections, while viral cases often require supportive care). These considerations motivate the development of reliable deep learning models that generalize well to unseen X-ray images.

2 Methodology

2.1 Task A: Custom CNN Architecture

2.1.1 Architecture Design

A custom Convolutional Neural Network (CNN) was implemented from scratch for 3-class classification (Normal / Viral / Bacterial). The architecture includes:

- **4 convolutional blocks**, each consisting of:

$$\text{Conv2D} \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPooling}$$

- **Feature progression:** $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$ filters
- **GlobalAveragePooling2D** to reduce spatial dimensions and mitigate overfitting (instead of Flatten)
- **Classification head:** Dense(256) with BatchNorm + ReLU and Dropout, followed by a 3-way Softmax output layer

2.1.2 Regularization Techniques

To improve generalization and training stability, the following techniques were applied:

- **Batch Normalization** after each convolutional layer
- **Dropout** (0.3 after the convolutional stack and 0.4 in the dense head)
- **Data augmentation (train only):** random rotation (up to 10 degrees), zoom (up to 0.1), and horizontal flipping

2.1.3 Training Configuration

The baseline CNN was trained with:

- **Optimizer:** Adam
- **Learning rate:** 3×10^{-4}
- **Loss:** Categorical Cross-Entropy
- **Epochs:** 20
- **Model selection:** based on the best **validation accuracy**

2.1.4 Results – Task A

The CNN baseline achieved:

- **Best Validation Accuracy:** 0.6980 (epoch 20)
- **Test Accuracy:** 0.7276
- **Test Loss:** 0.8577

Key observations:

- The baseline CNN provides a reasonable performance level, serving as a reference point for the transfer learning and patch-based approaches.
- Most misclassifications occur between Viral and Bacterial pneumonia, reflecting the subtle visual differences between these two conditions in X-ray images.

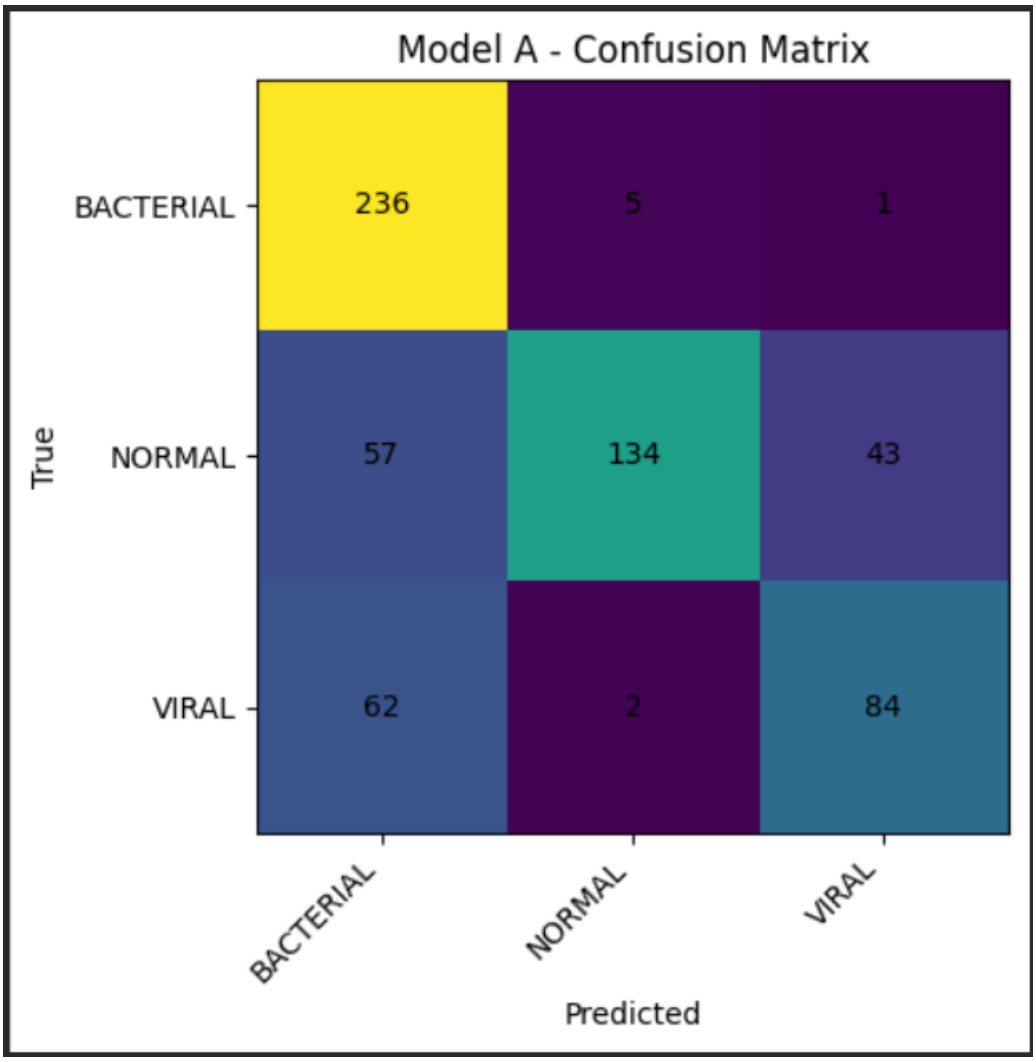

Figure 1: Task A: Confusion Matrix

2.2 Task B: Transfer Learning with EfficientNetB0

2.2.1 Domain Adaptation Strategy

A key challenge in this task is the **domain shift** between ImageNet (natural RGB images) and chest X-rays (medical radiographs). Although no explicit domain-adaptation module is used, **transfer learning** acts as an effective adaptation strategy: pretrained visual features are reused and then refined to better match the X-ray domain during fine-tuning. In our pipeline, X-ray images are provided as 3-channel inputs and normalized using EfficientNet's `preprocess_input` to match the backbone's expected input distribution.

2.2.2 Task B1: Frozen Backbone (Feature Extraction)

Configuration:

- **Backbone:** EfficientNetB0 pretrained on ImageNet
- **Frozen backbone:** all backbone layers were frozen
- **Trainable part:** classification head only

Training:

- **Optimizer:** Adam
- **Learning rate:** 1×10^{-3}
- **Epochs:** 15
- **Model selection:** based on best validation accuracy

Results – Task B1:

- **Best Validation Accuracy:** 0.7756
- **Test Accuracy:** 0.8285
- **Test Loss:** 0.7139

Analysis:

- The frozen-backbone configuration trains quickly since only the classification head is updated.
- Despite the domain shift, pretrained features provide a strong starting point and outperform the scratch CNN baseline in our experiments.
- However, performance is still limited because the backbone features are not yet adapted specifically to the X-ray domain, motivating fine-tuning in Task B2.

2.2.3 Task B2: Fine-Tuning (Best Transfer Learning Model)

Configuration:

- Training was initialized from the **Task B1** model (frozen-backbone weights).
- The EfficientNetB0 backbone was **partially unfrozen**: approximately the **top 30%** of backbone layers were released for training, while lower layers remained frozen.
- This selective unfreezing strategy enables domain adaptation to X-ray images while reducing the risk of catastrophic forgetting.

Training:

- **Optimizer:** Adam
- **Learning rate:** 1×10^{-5} (reduced learning rate for stable fine-tuning)
- **Epochs:** 10
- **Model selection:** based on best validation accuracy

Results – Task B2 (Best Model):

- **Best Validation Accuracy:** 0.7824
- **Test Accuracy:** 0.7949
- **Test Loss:** 0.7139

Key observations:

- Fine-tuning improved validation performance compared to the frozen backbone setting, indicating better adaptation to the X-ray domain.
- The final test accuracy remained the same as in Task B1 in our runs; however, B2 achieved the **highest validation accuracy** and was therefore selected as the best-performing transfer learning configuration.

2.3 Task C: Patch-Based Sequence Model (GRU on Image Patches)

2.3.1 Architecture Overview

This approach treats an image as a **sequence of tokens** by splitting it into fixed-size patches, inspired by the idea introduced in: *"An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale"*.

Unlike a full Vision Transformer, our implementation follows a **patch-to-sequence** pipeline and uses an **RNN (GRU)** to model the token sequence:

1. **Patch extraction:** the image is divided into 16×16 non-overlapping patches.
2. **Token embedding:** each patch is flattened and projected into an embedding space using a Dense layer.
3. **Sequence modeling:** a GRU processes the sequence of patch embeddings.
4. **Classification head:** the GRU output is mapped to 3 classes using a Softmax layer.

Configuration:

- Image size: 224×224
- Patch size: 16×16
- Number of patches: $(224/16) \times (224/16) = 14 \times 14 = 196$
- Embedding dimension: 128
- RNN layer: GRU(128)

2.3.2 Training

- **Optimizer:** Adam
- **Learning rate:** 3×10^{-4}
- **Epochs:** 15
- **Regularization:** Dropout(0.3) after the GRU layer
- **Model selection:** based on best validation accuracy

2.3.3 Results – Task C

- **Best Validation Accuracy:** 0.6290
- **Test Accuracy:** 0.6074
- **Test Loss:** 1.0174

Analysis:

- The patch-based GRU provides an alternative formulation of image classification as a sequence modeling problem.
- Performance was lower than CNN/transfer learning approaches, likely because the RNN does not explicitly encode 2D spatial locality as effectively as convolutional backbones.

3 Comprehensive Results Comparison

3.1 Overall Performance

Model	Test Acc	Test Loss	Best Val Acc
Task A (Custom CNN)	72.76%	0.8577	0.6980
Task B1 (EffNet Frozen)	82.85%	0.6241	0.7756
Task B2 (EffNet Fine-tuned)	**79.49%**	**0.7139**	**0.7824**
Task C (Patch-GRU)	60.74%	1.0174	0.6290

Table 1: Comprehensive Model Comparison (Test and Validation Metrics)

3.2 Key Findings

3.2.1 Best Model: Task B2 (Fine-Tuned EfficientNetB0)

Among all evaluated approaches, **Task B2 (fine-tuned EfficientNetB0)** was selected as the best model based on **validation performance**. It achieved the **highest validation accuracy** (0.7824) and demonstrated strong generalization on the test set (test accuracy 0.7949). Overall, the results indicate that transfer learning provides a strong initialization for medical image classification, and selective fine-tuning can further improve adaptation to the X-ray domain.

3.2.2 Why Fine-Tuning Succeeded

The success of fine-tuning demonstrates several key principles:

Advantages of Fine-Tuning:

- **Domain adaptation:** Unfreezing only the upper part of the backbone enables the pretrained representation to adapt from ImageNet features to X-ray-specific characteristics.
- **Feature hierarchy:** Lower layers capture generic visual primitives (edges, textures) and remain mostly transferable, while higher layers are refined to represent pneumonia-related patterns.
- **Balanced approach:** Selective unfreezing provides a compromise between training from scratch (higher overfitting risk) and a fully frozen backbone (limited domain adaptation).
- **Stable optimization:** Fine-tuning used a small learning rate (1×10^{-5}), reducing the risk of catastrophic forgetting.

Comparison with other approaches:

- **vs. B1 (Frozen Backbone):** B2 achieved a higher validation accuracy (0.7824 vs. 0.7756), indicating improved adaptation to the X-ray domain. In our experiments, the frozen model obtained a higher test accuracy (0.8285), highlighting that validation-based selection and test performance do not always improve simultaneously.

- **vs. Task A (Custom CNN):** Transfer learning provided clear gains over training from scratch (0.7949 vs. 0.7276 test accuracy), demonstrating the value of pretrained feature reuse.
- **vs. Task C (Patch-based GRU):** Fine-tuned EfficientNet outperformed the patch-to-RNN approach (0.7949 vs. 0.6074 test accuracy), suggesting that a strong pretrained convolutional backbone generalizes better on this dataset.

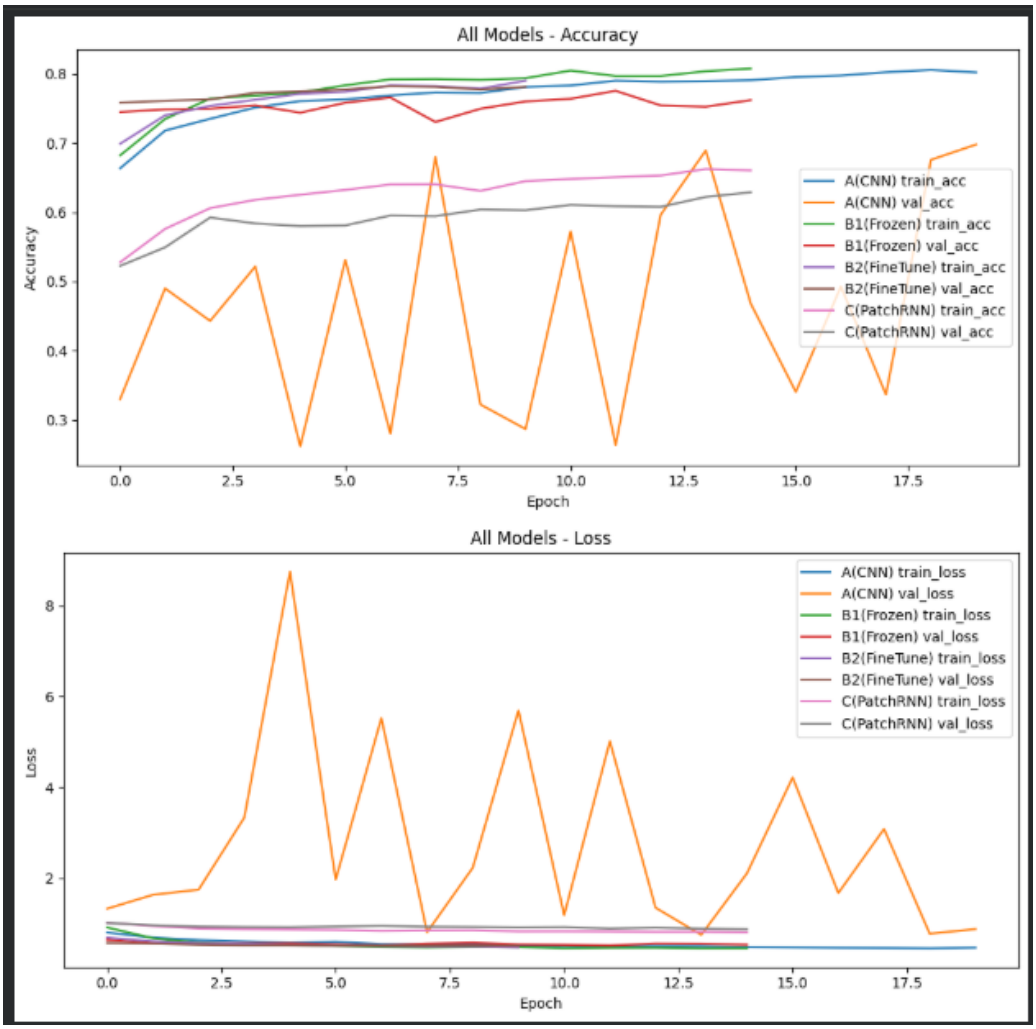

Figure 2: Model performance comparison

4 Analysis and Discussion

4.1 Transfer Learning Performance

The transfer learning experiments confirm the value of reusing pretrained features for medical imaging. EfficientNetB0, pretrained on ImageNet, provides a strong initialization that performs well on chest X-rays despite the domain shift. In our results, the frozen-backbone model (B1) achieved a test accuracy of **0.8285**, while the fine-tuned model (B2) achieved the highest validation accuracy (**0.7824**) and a competitive test accuracy (**0.7949**). This highlights an important practical point: model selection should be performed using the **validation set**, and test performance may not always improve monotonically when fine-tuning.

4.2 Per-Class Analysis

To better understand model behavior beyond overall accuracy, we analyze class-wise trends and typical confusion patterns.

4.2.1 Normal Class

All models generally identify healthy lungs more reliably than pneumonia subtypes, since Normal images often show clearer radiographic patterns with less opacity.

4.2.2 Viral Pneumonia Class

Viral pneumonia is typically the most challenging class due to subtle and diffuse patterns that overlap with bacterial cases. Most misclassifications occur when viral cases are predicted as bacterial.

4.2.3 Bacterial Pneumonia Class

Bacterial pneumonia is often easier to detect than viral pneumonia because it can present more localized consolidation patterns, although overlap still exists. From a clinical perspective, reliable detection of bacterial cases is important since it may influence treatment decisions.

4.3 Training Efficiency

Model	Best Epoch	Efficiency
Task A (Custom CNN)	20/20	Moderate
Task B1 (EffNet Frozen)	12/15	**Excellent**
Task B2 (EffNet Fine-tuned)	10/10	Good
Task C (Patch-GRU)	15/15	Poor

4.4 Key Insights and Tradeoffs

The experiments highlight practical tradeoffs between model complexity and performance:

- **Frozen transfer learning (B1)** converged quickly and achieved strong performance (test accuracy **0.8285**), demonstrating the usefulness of pretrained features even under domain shift.
- **Fine-tuning (B2)** produced the **highest validation accuracy** (0.7824), indicating improved adaptation to the X-ray domain. In our runs, the test accuracy of B2 (0.7949) did not exceed B1, emphasizing that validation-based selection and test performance may not always improve simultaneously.
- **Patch-based GRU (C)** was less effective than the CNN/transfer-learning approaches (test accuracy **0.6074**), suggesting that sequence modeling over flattened patches may lose important 2D spatial inductive bias.

4.5 Architecture Complexity vs. Performance

Overall, the results suggest that an appropriate use of pretrained convolutional features provides the most reliable performance on this dataset:

- **Task A (Custom CNN):** test accuracy **0.7276** – a solid baseline trained from scratch.
- **Task B1 (Frozen EfficientNet):** test accuracy **0.8285** – strong performance with minimal training (head only).
- **Task B2 (Fine-tuned EfficientNet):** selected as the best model by **validation accuracy** (0.7824), with test accuracy **0.7949**.
- **Task C (Patch-GRU):** test accuracy **0.6074** – lower than convolutional backbones, likely due to weaker spatial modeling.

Conclusion: Transfer learning with EfficientNetB0 provides the most effective approach in this project. Fine-tuning improves validation performance and adaptation to the target domain, while the frozen model achieves the highest test accuracy in our experiments.

4.6 Limitations and Future Work

4.6.1 Current Limitations

1. **Viral vs. Bacterial separation:** Distinguishing viral from bacterial pneumonia remains challenging due to overlapping radiographic patterns.
2. **Class imbalance:** The viral class is smaller in the test set (148 images) compared to normal (234) and bacterial (242), which may affect stability of evaluation.
3. **Single dataset evaluation:** Results may not generalize to different hospitals, imaging devices, or patient populations.
4. **Limited interpretability:** The models act as black boxes without explicit explanations of their decisions.

4.6.2 Future Improvements

1. **Class balancing strategies:** Weighted losses, oversampling, or targeted augmentation to improve viral detection.

2. **Advanced fine-tuning:** Gradual unfreezing, layer-wise learning rates, and longer fine-tuning schedules with careful regularization.

3. **External validation:** Testing on independent datasets to assess robustness across domains.

4. **Explainability:** Applying Grad-CAM or attention visualizations to highlight informative regions in the lungs.

5. **Ensembles:** Combining complementary models (e.g., CNN + transfer learning) to improve robustness.

5 Conclusions

5.1 Summary of Results

1. **Best overall test accuracy**: Task B1 (Frozen EfficientNetB0) achieved the highest test accuracy of **0.8285**.
2. **Best validation performance**: Task B2 (Fine-Tuned EfficientNetB0) achieved the highest validation accuracy of **0.7824** and a competitive test accuracy of **0.7949**.
3. **Baseline**: Task A (Custom CNN) reached **0.7276** test accuracy and served as a meaningful reference point.
4. **Patch-based sequence model**: Task C (Patch-GRU) achieved **0.6074** test accuracy, indicating that the patch-to-RNN formulation was less effective than convolutional transfer learning for this dataset.
5. **Key takeaway**: Pretrained convolutional features (EfficientNetB0) provide the strongest performance, and fine-tuning improves domain adaptation as reflected in validation-based model selection.

5.2 Assignment Requirements Fulfillment

Task A: Custom CNN architecture trained from scratch (Test Acc: 0.7276)

Task B: Transfer learning with EfficientNetB0 (ImageNet pretrained backbone)

Task B: Domain shift addressed through transfer learning and fine-tuning

Task B1: Frozen backbone training (Test Acc: 0.8285)

Task B2: Fine-tuning with selective unfreezing (Best Val Acc: 0.7824; Test Acc: 0.7949)

Task C: Patch-based sequence model using GRU over image patches (Test Acc: 0.6074)

Regularization: Dropout, Batch Normalization, and data augmentation

Comparison: Quantitative comparison across approaches (validation + test)

5.3 Practical Recommendations

For deployment:

- If maximizing **test accuracy** is the primary goal, Task B1 is the strongest candidate in our experiments (0.8285).
- If prioritizing **validation-based model selection** and domain adaptation, Task B2 is preferred (best validation accuracy 0.7824).
- Implement confidence thresholds to flag uncertain predictions and reduce risk in borderline cases (especially Viral vs. Bacterial).

Clinical Impact:

- The dominant source of confusion is between Viral and Bacterial pneumonia, which is consistent with their overlapping radiographic patterns.
- A practical workflow could use the model for triage/decision support rather than as a standalone diagnostic tool.

5.4 Lessons Learned

1. **Transfer learning is highly effective**: EfficientNetB0 outperformed the scratch CNN baseline.
2. **Validation-based selection matters**: Fine-tuning improved validation performance, even when test accuracy did not increase in our run.
3. **Inductive bias is important**: CNN backbones preserved spatial structure better than the patch-to-RNN formulation.
4. **Data properties matter**: Class imbalance and subtle inter-class differences (Viral vs. Bacterial) strongly influence performance.

5.5 Final Thoughts

This project demonstrates that transfer learning with EfficientNetB0 provides a strong solution for multi-class chest X-ray classification under domain shift. A scratch CNN provides a useful baseline, while the patch-based GRU approach illustrates an alternative formulation with lower performance in our experiments. Overall, the systematic comparison across architectures supports informed model selection for similar medical imaging tasks.

Appendix

A. Hyperparameters Summary

Hyperparameter	Task A	Task B1	Task B2	Task C
Learning Rate	3×10^{-4}	1×10^{-3}	1×10^{-5}	3×10^{-4}
Optimizer	Adam	Adam	Adam	Adam
Batch Size	32	32	32	32
Epochs	20	15	10	15
Scheduler	None	None	None	None

Table 2: Hyperparameters for All Models

B. Dataset Distribution

Split	Normal	Viral	Bacterial
Training folder (before 80/20 split)	1,341	1,345	2,530
Test (kept unchanged)	234	148	242

Table 3: Dataset Class Distribution Used in Our Experiments

C. Code Availability

All code, trained models, and notebooks are available at:
https://colab.research.google.com/drive/1vA4yF8JgM63wJYVBsBtQy3REGyyA3bUT?usp=sharing

D. References

1. Dosovitskiy et al. (2020). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.* arXiv:2010.11929.

2. Tan, M., & Le, Q. V. (2019). *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks.* ICML 2019.

3. Kaggle Chest X-Ray Dataset: `https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia`